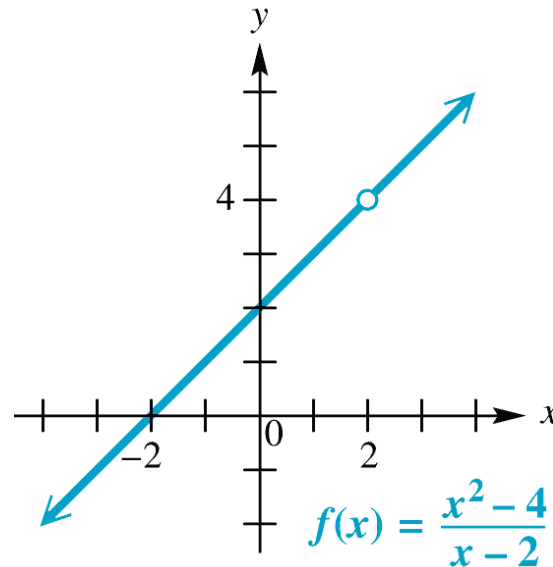


Limit of a Function

The function

$$f(x) = \frac{x^2 - 4}{x - 2}$$



is not defined at $x = 2$, so its graph has a “hole” at $x = 2$.

Limit of a Function

Values of $f(x) = \frac{x^2 - 4}{x - 2}$ may be computed near $x = 2$

x approaches 2

x	1.9	1.99	1.999→	←2.001	2.01	2.1
$f(x)$	3.9	3.99	3.999→	←4.001	4.01	4.1

$f(x)$ approaches 4

Limit of a Function

The values of $f(x)$ get closer and closer to 4 as x gets closer and closer to 2.

We say that

“the limit of $\frac{x^2 - 4}{x - 2}$ as x approaches 2 equals 4”

and write

$$\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = 4.$$

Limit of a Function

Limit of a Function

Let f be a function and let a and L be real numbers.

L is the limit of $f(x)$ as x approaches a , written

$$\lim_{x \rightarrow a} f(x) = L,$$

if the following conditions are met.

1. As x assumes values closer and closer (but not equal) to a on both sides of a , the corresponding values of $f(x)$ get closer and closer (and are perhaps equal) to L .
2. The value of $f(x)$ can be made as close to L as desired by taking values of x arbitrarily close to a .

Finding the Limit of a Polynomial Function

Example Find $\lim_{x \rightarrow 1} (x^2 - 3x + 4)$.

Solution The behavior of $f(x) = x^2 - 3x + 4$ near $x = 1$ can be determined from a table of values,

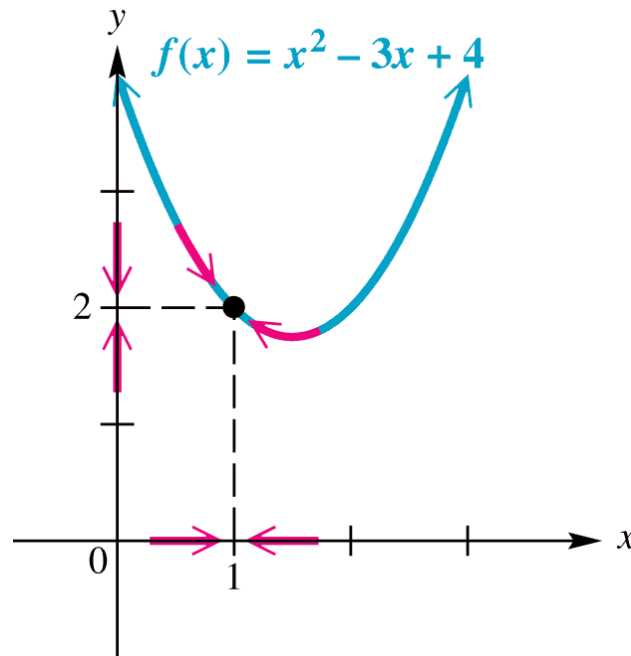
x approaches 1

x	.9	.99	.999 $\rightarrow$	$\leftarrow$ 1.001	1.01	1.1
$f(x)$	2.11	2.0101	2.001 $\rightarrow$	$\leftarrow$ 1.999	1.9901	1.91

$f(x)$ approaches 2

Finding the Limit of a Polynomial Function

Solution or from a graph of $f(x)$.



We see that

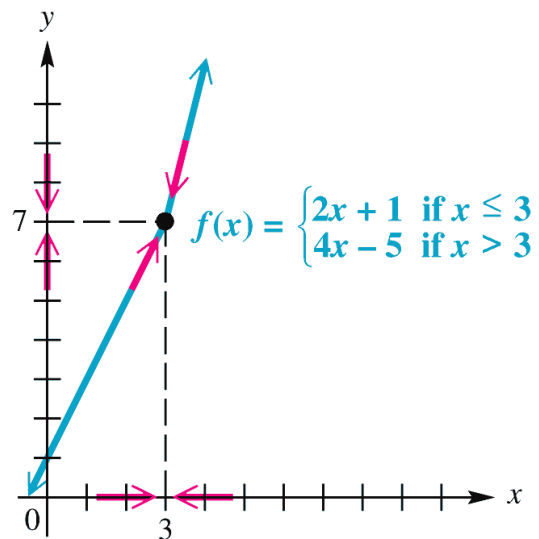
$$\lim_{x \rightarrow 1} (x^2 - 3x + 4) = 2.$$

Finding the Limit of a Polynomial Function

Example Find $\lim_{x \rightarrow 3} f(x)$ where

$$f(x) = \begin{cases} 2x + 1 & \text{if } x \leq 3 \\ 4x - 5 & \text{if } x > 3 \end{cases}.$$

Solution Create a graph and table.



Finding the Limit of a Polynomial Function

Solution

x approaches 3

x	2.9	2.99	2.999→	←3.001	3.01	3.1
$f(x)$	6.8	6.98	6.998→	←7.004	7.04	7.4

$f(x)$ approaches 7

Therefore

$$\lim_{x \rightarrow 3} f(x) = 7.$$

Limits That Do Not Exist

- If there is no single value that is approached by $f(x)$ as x approaches a , we say that $f(x)$ does not have a limit as x approaches a , or $\lim_{x \rightarrow 2} f(x)$ does not exist.

Determining Whether a Limit Exists

Example Find $\lim_{x \rightarrow 2} f(x)$ where

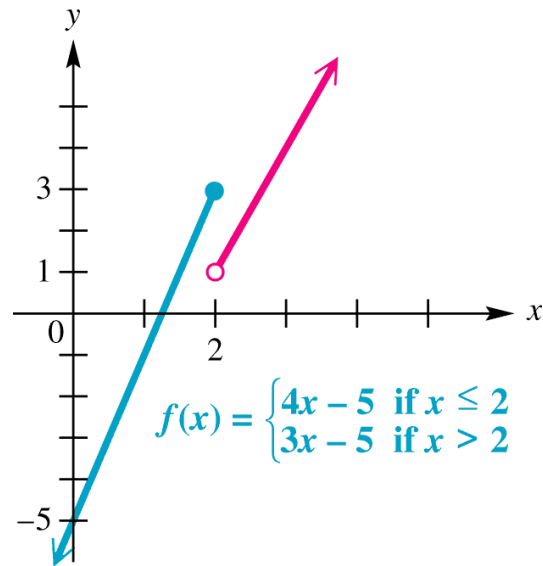
$$f(x) = \begin{cases} 4x - 5 & \text{if } x \leq 2 \\ 3x - 5 & \text{if } x > 2 \end{cases}$$

Solution Construct a table and graph

x	1.9	1.99	$1.999 \rightarrow$	$\leftarrow 2.001$	2.01	2.1
$f(x)$	2.6	2.96	$2.996 \rightarrow$	$\leftarrow 1.003$	1.03	1.3

Determining Whether a Limit Exists

Solution



$f(x)$ approaches 3 as x gets closer to 2 from the left,
 $f(x)$ approaches 1 as x gets closer to 2 from the right.

Therefore, $\lim_{x \rightarrow 2} f(x)$ does not exist.

Determining Whether a Limit Exists

Example Find $\lim_{x \rightarrow 0} f(x)$ where $f(x) = \frac{1}{x^2}$.

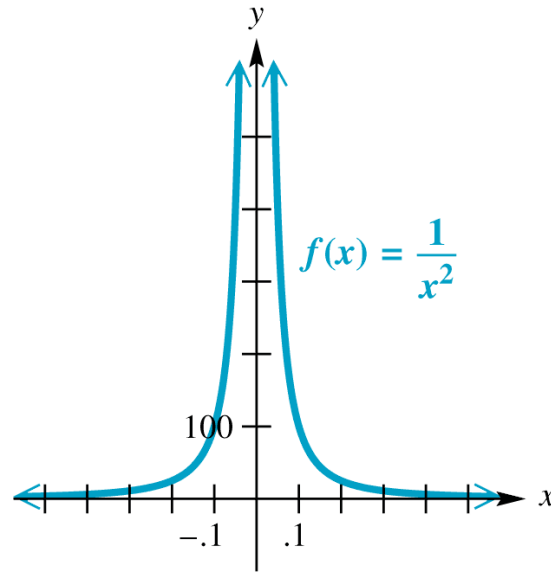
Solution Construct a table and graph

x	-.1	-.01	-.001 $\rightarrow$
$f(x)$	100	10,000	1,000,000 $\rightarrow$

x	$\leftarrow$.001	.01	.1
$f(x)$	$\leftarrow$ 1,000,000	10,000	100

Determining Whether a Limit Exists

Solution



As x approaches 0, the corresponding values of $f(x)$ grow arbitrarily large.

Therefore, $\lim_{x \rightarrow 0} \frac{1}{x^2}$ does not exist.

Limit of a Function

Conditions Under Which $\lim_{x \rightarrow a} f(x)$ Fails To Exist

1. $f(x)$ approaches a number L as x approaches a from the left and $f(x)$ approaches a different number M as x approaches a from the right.
2. $f(x)$ becomes infinitely large in absolute value as x approaches a from either side.
3. $f(x)$ oscillates infinitely many times between two fixed values as x approaches a .

Limits

1. An Introduction To Limits
- 2. Techniques for Calculating Limits**
3. One-Sided Limits; Limits Involving Infinity

Techniques For Calculating Limits

Rules for Limits

1. **Constant rule** If k is a constant real number, $\lim_{x \rightarrow a} k = k$.

2. **Limit of x rule** $\lim_{x \rightarrow a} x = a$.

For the following rules, we assume that $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$

both exist

3. **Sum and difference rules**

$$\lim_{x \rightarrow a} [f(x) \pm g(x)] = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x).$$

Techniques For Calculating Limits

Rules for Limits

4. Product Rule

$$\lim_{x \rightarrow a} [f(x) \cdot g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x).$$

5. Quotient Rule

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)}.$$

provided $\lim_{x \rightarrow a} g(x) \neq 0$.

Finding a Limit of a Linear Function

Example Find $\lim_{x \rightarrow 4} (3 + 2x)$.

Solution

$$\begin{aligned}\lim_{x \rightarrow 4} (3 + 2x) &= \lim_{x \rightarrow 4} 3 + \lim_{x \rightarrow 4} 2x \\ &= 3 + \lim_{x \rightarrow 4} 2 \cdot \lim_{x \rightarrow 4} x \quad \text{Rules 1 and 4} \\ &= 3 + 2 \cdot 4 \quad \text{Rules 1 and 2} \\ &= 11\end{aligned}$$

Finding a Limit of a Polynomial Function with One Term

Example Find $\lim_{x \rightarrow 5} 3x^2$.

Solution

$$\begin{aligned}\lim_{x \rightarrow 5} 3x^2 &= \lim_{x \rightarrow 5} 3 \cdot \lim_{x \rightarrow 5} x^2 && \text{Rule 4} \\ &= 3 \cdot \lim_{x \rightarrow 5} x^2 && \text{Rule 1} \\ &= 3 \cdot \lim_{x \rightarrow 5} x \cdot \lim_{x \rightarrow 5} x && \text{Rule 4} \\ &= 3 \cdot 5 \cdot 5 && \text{Rule 2} \\ &= 75\end{aligned}$$

Finding a Limit of a Polynomial Function with One Term

For any polynomial function in the form $f(x) = kx^n$,

$$\lim_{x \rightarrow a} f(x) = k \cdot a^n = f(a).$$

Finding a Limit of a Polynomial Function

Example Find $\lim_{x \rightarrow 2} (4x^3 - 6x + 1)$.

Solution $\lim_{x \rightarrow 2} (4x^3 - 6x + 1) = \lim_{x \rightarrow 2} 4x^3 - \lim_{x \rightarrow 2} 6x + \lim_{x \rightarrow 2} 1$

Rule 3

$$= 4 \cdot 2^3 - 6 \cdot 2 + 1$$

$$= 21$$

Techniques For Calculating Limits

Rules for Limits (Continued)

For the following rules, we assume that $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ both exist.

6. Polynomial rule If $p(x)$ defines a polynomial function, then

$$\lim_{x \rightarrow a} p(x) = p(a).$$

Techniques For Calculating Limits

Rules for Limits (Continued)

7. Rational function rule If $f(x)$ defines a rational function $\frac{p(x)}{q(x)}$ with $q(a) \neq 0$ then

$$\lim_{x \rightarrow a} f(x) = f(a).$$

8. Equal functions rule If $f(x) = g(x)$ for all $x \neq a$, then

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x).$$

Techniques For Calculating Limits

Rules for Limits (Continued)

9. Power rule For any real number k ,

$$\lim_{x \rightarrow a} [f(x)]^k = \left[\lim_{x \rightarrow a} f(x) \right]^k$$

provided this limit exists.

Techniques For Calculating Limits

Rules for Limits (Continued)

10. Exponent rule For any real number $b > 0$,

$$\lim_{x \rightarrow a} b^{f(x)} = b^{\lim_{x \rightarrow a} f(x)}.$$

11. Logarithm rule For any real number $b > 0$ with $b \neq 1$,

$$\lim_{x \rightarrow a} [\log_b f(x)] = \log_b \left[\lim_{x \rightarrow a} f(x) \right]$$

provided that $\lim_{x \rightarrow a} f(x) > 0$.

Finding a Limit of a Rational Function

Example Find $\lim_{x \rightarrow 1} \frac{x^2 + 2x - 3}{x^2 - 3x + 2}$.

Solution Rule 7 cannot be applied directly since the denominator is 0. First factor the numerator and denominator

$$\frac{x^2 + 2x - 3}{x^2 - 3x + 2} = \frac{(x + 3)(x - 1)}{(x - 2)(x - 1)} = \frac{x + 3}{x - 2}$$

Finding a Limit of a Rational Function

Solution Now apply Rule 8 with

$$f(x) = \frac{x^2 + 2x - 3}{x^2 - 3x + 2} \quad \text{and} \quad g(x) = \frac{x + 3}{x - 2}$$

so that $f(x) = g(x)$ for all $x \neq 1$.

Finding a Limit of a Rational Function

Solution $\lim_{x \rightarrow 1} \frac{x^2 + 2x - 3}{x^2 - 3x + 2} = \lim_{x \rightarrow 1} \frac{x + 3}{x - 2}$ Rule 8

$$= \frac{1 + 3}{1 - 2} \quad \text{Rule 6}$$

$$= -4$$

Limits

- 1 An Introduction To Limits
- 2 Techniques for Calculating Limits
- 3 **One-Sided Limits; Limits Involving Infinity**

One-Sided Limits

Limits of the form

$$\lim_{x \rightarrow a} f(x) = L$$

are called **two-sided limits** since the values of x get close to a from both the right and left sides of a .

Limits which consider values of x on only one side of a are called **one-sided limits**.

One-Sided Limits

The **right-hand limit**,

$$\lim_{x \rightarrow a^+} f(x) = L$$

is read “the limit of $f(x)$ as x approaches a from the right is L .”

As x gets closer and closer to a from the right ($x > a$), the values of $f(x)$ get closer and closer to L .

One-Sided Limits

The **left-hand limit**,

$$\lim_{x \rightarrow a^-} f(x) = L$$

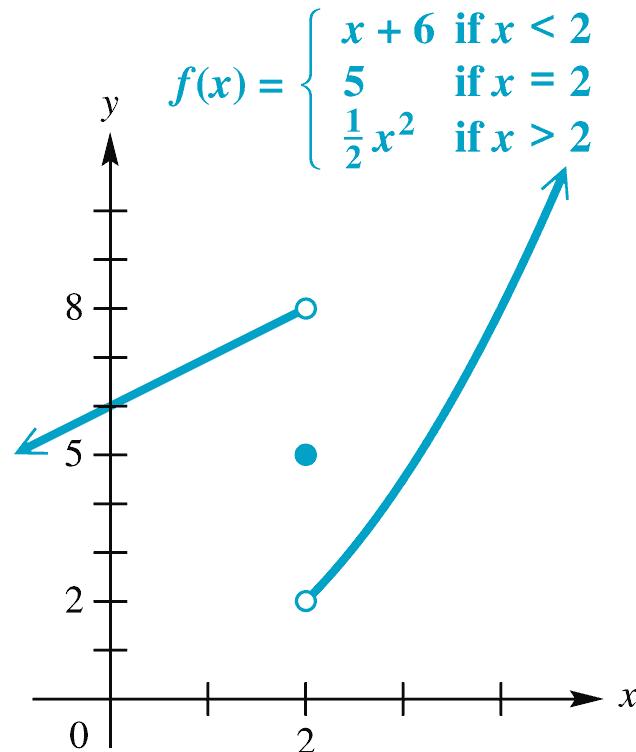
is read “the limit of $f(x)$ as x approaches a from the left is L .”

As x gets closer and closer to a from the right ($x < a$), the values of $f(x)$ get closer and closer to L .

Finding One-Sided Limits of a Piecewise-Defined Function

Example Find $\lim_{x \rightarrow 2^+} f(x)$ and $\lim_{x \rightarrow 2^-} f(x)$ where

$$f(x) = \begin{cases} x + 6 & \text{if } x < 2 \\ 5 & \text{if } x = 2 \\ \frac{1}{2}x^2 & \text{if } x > 2 \end{cases}$$



Finding One-Sided Limits of a Piecewise-Defined Function

Solution Since $x > 2$ in $\lim_{x \rightarrow 2^+} f(x)$ use the formula

$f(x) = \frac{1}{2}x^2$. In the limit $\lim_{x \rightarrow 2^-} f(x)$, where $x < 2$, use

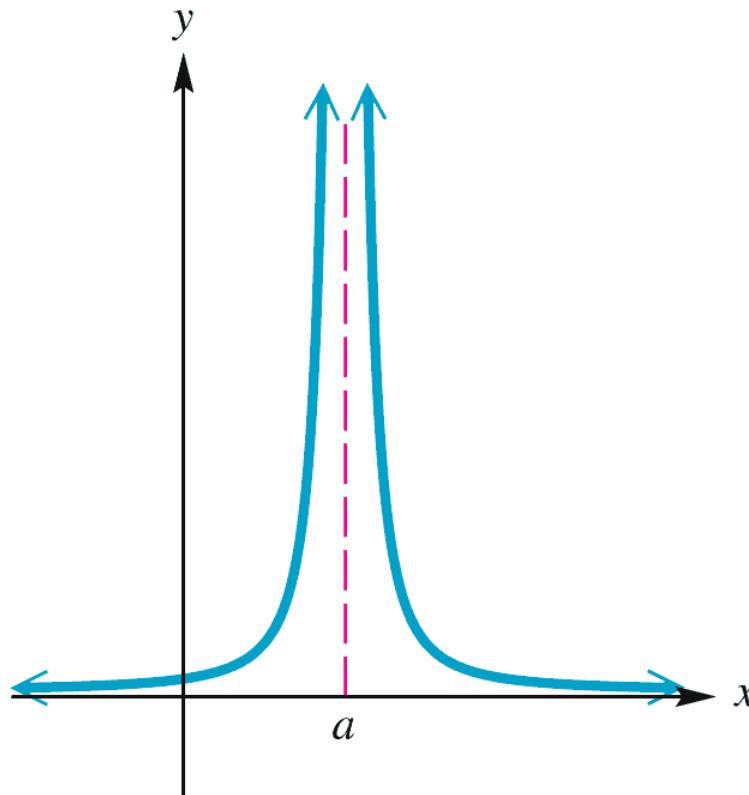
$$f(x) = x + 6.$$

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} \frac{1}{2}x^2 = \frac{1}{2}(2^2) = 2$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} (x + 6) = 2 + 6 = 8$$

Infinity as a Limit

A function may increase without bound as x gets closer and closer to a from the right



Infinity as a Limit

The right-hand limit does not exist but the behavior is described by writing

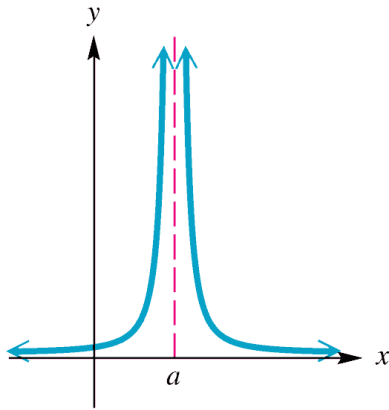
$$\lim_{x \rightarrow a^+} f(x) = \infty$$

If the values of $f(x)$ decrease without bound, write

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

The notation is similar for left-handed limits.

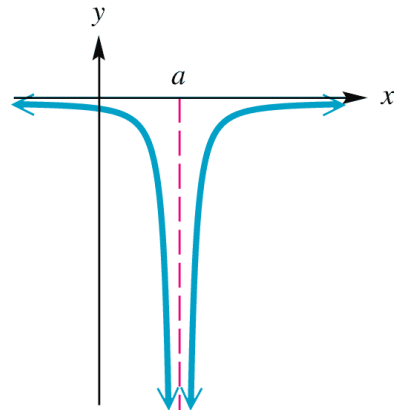
Infinity as a Limit



$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^+} f(x) = \infty$$

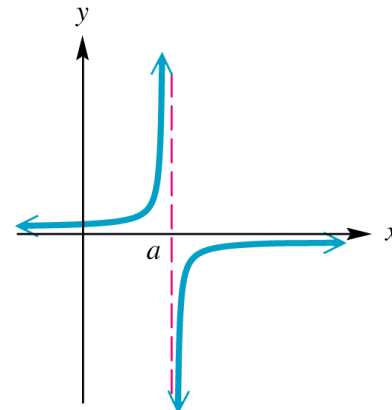
$$\lim_{x \rightarrow a} f(x) = \infty$$



$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

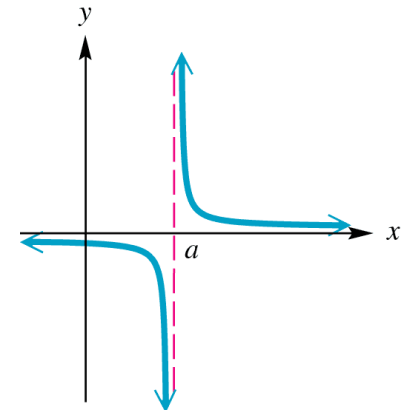
$$\lim_{x \rightarrow a} f(x) = -\infty$$



$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

$$\lim_{x \rightarrow a} f(x) \text{ does not exist.}$$



$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a} f(x) \text{ does not exist.}$$

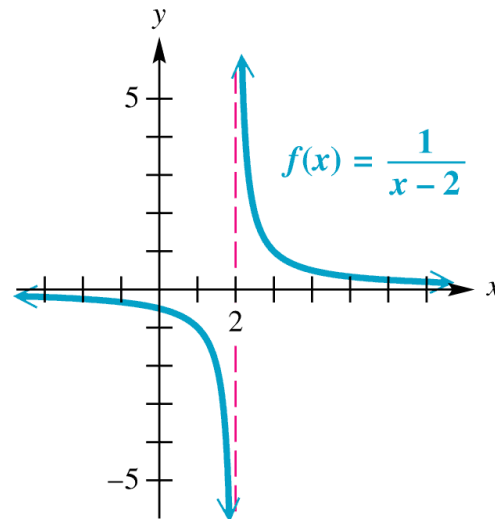
Summary of infinite limits

Finding One-Sided Limits

Example Find $\lim_{x \rightarrow 2^+} f(x)$ and $\lim_{x \rightarrow 2^-} f(x)$ where

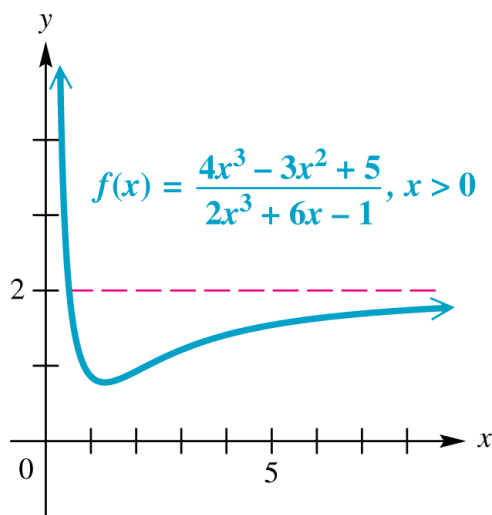
$$f(x) = \frac{1}{x-2}.$$

Solution From the graph

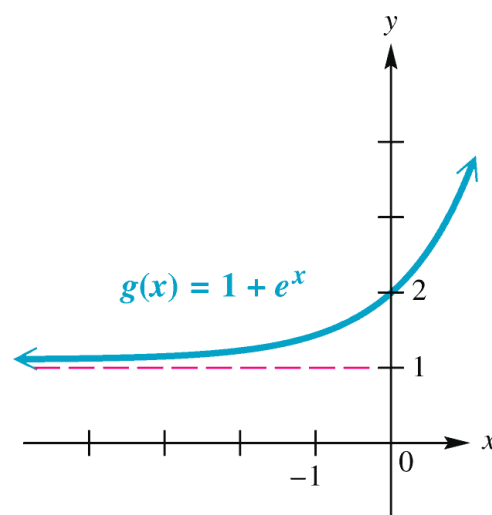


Limits as x Approaches $\pm\infty$

A function may approach an asymptotic value as x moves in the positive or negative direction.



$$\lim_{x \rightarrow \infty} f(x) = 2$$



$$\lim_{x \rightarrow -\infty} g(x) = 1$$

Limits as x Approaches $\pm\infty$

The notation,

$$\lim_{x \rightarrow \infty} f(x) = L$$

is read “the limit of $f(x)$ as x approaches infinity is L .”

The values of $f(x)$ get closer and closer to L as x gets larger and larger.

Limits as x Approaches $\pm\infty$

The notation,

$$\lim_{x \rightarrow -\infty} f(x) = L$$

is read “the limit of $f(x)$ as x approaches negative infinity is L .”

The values of $f(x)$ get closer and closer to L as x assumes negative values of larger and larger magnitude.

Finding Limits at Infinity

Example Find $\lim_{x \rightarrow \infty} f(x)$ and $\lim_{x \rightarrow -\infty} f(x)$ where

$$f(x) = 5 + \frac{10}{1 + e^{-.25x}}.$$

Solution As $x \rightarrow \infty$ the values of $e^{-.25x}$ get arbitrarily close to 0 so

$$\lim_{x \rightarrow \infty} f(x) = 5 + \frac{10}{1 + 0} = 15.$$

Limits as x Approaches $\pm\infty$

Limits at infinity of $\frac{1}{x^n}$

For any positive real number n ,

$$\lim_{x \rightarrow \infty} \frac{1}{x^n} = 0 \quad \text{and} \quad \lim_{x \rightarrow -\infty} \frac{1}{x^n} = 0.$$

Finding a Limit at Infinity

Example Find $\lim_{x \rightarrow \infty} \frac{5x^2 - 7x + 1}{2x^2 + x + 5}$.

Solution Divide numerator and denominator by the highest power of x involved, x^2 .

$$\lim_{x \rightarrow \infty} \frac{5x^2 - 7x + 1}{2x^2 + x + 5} = \lim_{x \rightarrow \infty} \frac{5 - \frac{7}{x} + \frac{1}{x^2}}{2 + \frac{1}{x} + \frac{5}{x^2}}$$

Finding a Limit at Infinity

Solution

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{5x^2 - 7x + 1}{2x^2 + x + 5} &= \lim_{x \rightarrow \infty} \frac{5 - \frac{7}{x} + \frac{1}{x^2}}{2 + \frac{1}{x} + \frac{5}{x^2}} \\ &= \frac{\lim_{x \rightarrow \infty} 5 - \frac{7}{x} + \frac{1}{x^2}}{\lim_{x \rightarrow \infty} 2 + \frac{1}{x} + \frac{5}{x^2}}\end{aligned}$$

Finding a Limit at Infinity

Solution

$$\begin{aligned}\lim_{x \rightarrow \infty} \frac{5x^2 - 7x + 1}{2x^2 + x + 5} &= \frac{\lim_{x \rightarrow \infty} 5 - 7 \cdot \lim_{x \rightarrow \infty} \frac{1}{x} + \lim_{x \rightarrow \infty} \frac{1}{x^2}}{\lim_{x \rightarrow \infty} 2 + \lim_{x \rightarrow \infty} \frac{1}{x} + 5 \cdot \lim_{x \rightarrow \infty} \frac{1}{x^2}} \\ &= \frac{5 - 0 + 0}{2 + 0 + 0} = \frac{5}{2}\end{aligned}$$